12 A metal block X of mass m, specific heat capacity c and temperature 80 °C is placed in good thermal contact with a second metal block Y of mass $2m$, specific heat capacity $2c$ and temperature 30 °C.

Assume no energy losses to the surroundings.

What will be the final temperature of both blocks?

A 30 °C **B** 40 °C **C** 55 °C **D** 70 °C

Ans: **B**

Heat lost by X = Heat gain by Y

$$mc(80 - T) = 2m(2c)(T - 30)$$

$$80 - T = 4T - 120$$

$$T = 40\ ^{o}C$$

13 A particle of a mass of 90.0 g undergoes simple harmonic motion. The graph in Fig. 13 shows